EXAM & VIVA DEFENSE STUDY MASTER GUIDE

# CareerForge AI: The Complete Architecture, Workflow & Viva Defense Manual

Everything You Need to Know to Ace Your Project Examination, Viva Voce & Technical Defense

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| **Project Title:** | CareerForge AI — Autonomous Career Readiness & Placement Platform |
| **Portals Developed:** | 1. Student Learning & Assessment Portal (index.html)<br>2. Faculty Administration & Placement Governance Portal (admin.html) |
| **Key Technologies:** | Vanilla JS SPA (Reactive Router), Supabase (PostgreSQL + Auth), Google Gemini 3.6 Flash AI Engine, HTML5 3D Flight Canvas |
| **Core Integrations:** | 1. Gemini API (Technical Drills, HR Drills, AI Mock Evaluation)<br>2. Supabase Cloud (Live Auth, student_states real-time sync)<br>3. Local Curated 25 Domain 16:9 Workstation Images |
| **Purpose of Guide:** | Quick revision of core logic, state flow, database models, examiner Q&A;, and live execution steps for examination day. |

**How to use this guide:** Read Section 1 for the 30-second project pitch. Section 2 explains how the Student Portal works step-by-step. Section 3 covers the Admin Portal. Section 4 reveals how APIs and Databases are connected. Section 5 provides direct, high-scoring answers to the most common viva questions examiners ask.

# 1. Project Summary & 30-Second Elevator Pitch

### If an examiner asks: 'What is your project about?'

*"CareerForge AI is an integrated, dual-portal career guidance and placement acceleration platform designed for engineering institutions. It solves the critical disconnect between collegiate curriculum and real-world hiring standards. For students, it provides psychometric diagnostic discovery across 25 industry domains, an interactive 3D resume builder, ATS resume analysis, domain-tailored technical and HR interview practice, and an AI Mock Interview evaluated by Google Gemini 3.6 Flash. For faculty and placement officers, it provides a real-time administration dashboard connected to Supabase PostgreSQL, enabling cohort analytics, detailed student dossier auditing, and one-click placement certificate endorsement."*

### Key Highlights to Mention to Examiners:

- **Dual-Portal Design:** Built with two distinct interfaces: Student Portal (index.html) and Faculty Admin Portal (admin.html).
- **Zero Dummy Data in Admin:** Admin portal queries real, live students from the Supabase student_states database.
- **Dual AI Roles:** Google Gemini 3.6 Flash acts as a *Principal Engineer* for Technical questions and a *Senior HR Director* for behavioral questions.
- **Anti-Gibberish Engine:** Strict client and server-side filters catch keyboard mash (e.g. 'sfdfdsfsnfbsjdfsdjf') giving 0% score with no unearned inflation.
- **100% Offline Asset Reliability:** 25 curated 16:9 4K workstation images load locally with 0ms lag and zero external API dependencies.
- **Readiness Index:** A composite algorithmic formula (0–100) reflecting a student's holistic readiness for placement drives.

# 2. Student Portal: Complete End-to-End Workflow

The Student Portal (index.html) takes a student through 5 progressive milestones:

| Step | Screen / Route | User Action | Behind the Scenes (How It Works) |
|---|---|---|---|
| 1 | #/splash<br>#/signup<br>#/login | Student registers or logs in with college email. | Supabase Auth validates credentials, creates user record in auth.users, issues JWT session token, and initializes empty state object in student_states table. |
| 2 | #/profile-setup | Enters name, roll number, department (CS/IT/ECE), and year. | Stores profile metadata into client state S.profile and persists to Supabase so the Faculty Admin can identify the student. |
| 3 | #/domains<br>#/assessment | Explores 25 career tracks; takes 12-question diagnostic test. | Assessment evaluates cognitive preferences, technical affinity, and work styles. Scores are calculated against a multi-attribute utility matrix across all 25 domains. |
| 4 | #/result<br>#/future | Views Top 5 domain matches with flowing glowing borders; opens 16:9 Workstation. | Renders matching percentages (e.g. 94% fit). Clicking a domain locks S.domain and displays the dedicated 16:9 workstation image from img/ instantly. |
| 5 | #/dashboard | Acts as central student headquarters. | Displays circular animated **Readiness Index (0-100)**, streak count, earned XP points, and shortcuts to practice modules. |
| 6 | #/resume<br>#/resume-builder | Analyzes resume or builds one using live 3D Warp Canvas. | ATS scanner simulates keyword matching, domain density, and quantifiable impact. 3D canvas projects 110 stars with type-reactive warp speed surge. |

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| **7** | #/interview-home<br>(Technical & HR) | Practices Technical & HR drills generated live by Gemini. | Calls gemini-3.6-flash via REST API. Gemini generates 5 fresh, domain-specific multiple choice questions with explanations and recruiter tips. |
| **8** | #/mock-home<br>(AI Mock Session) | Types free-form paragraph answers to real interview prompts. | Gemini evaluates answers. Pre-filters check for gibberish (0% score). Valid answers receive 4-tier score: Relevance, Communication, Technical Depth, and Structure. |
| **9** | #/reflection<br>#/rewards | Logs qualitative reflection; views unlocked trophy badges. | Awards completion XP points, updates daily streak, and unlocks milestone badges ('First Assessment', 'Interview Ace', etc.). |
| **10** | #/certificate<br>#/cert-preview | Applies for official placement readiness endorsement. | When Readiness Index is >= 75%, certificate unlocks. Status sets to 'Pending'. Once approved by Faculty Admin, renders official credential with verification hash. |

# 3. Faculty Admin Portal: Complete Workflow & Features

The Faculty Administration Portal (admin.html) provides full institutional control for Professors, Placement Officers (TPOs), and Academic HODs:

## A. Real-Time Institutional Dashboard & Analytics:

- **Live KPI Metrics:** Computes Total Registered Students, Average Institutional Readiness Score (mean across all students), Placement Eligible percentage (students >= 75%), and Pending Endorsements count.
- **Domain Enrollment Distribution:** Displays a visual distribution showing which careers students are targeting (e.g. 35% AI/ML, 25% Fullstack, 20% Cloud, 10% Cybersecurity, etc.).

## B. Live Student Cohort Table (Zero Dummy Data):

- **Real-Time Supabase Sync:** On page load, `admin.js` connects to the Supabase client and queries the `student_states` table. Only real registered students are displayed.
- **Multi-Criteria Search & Filtering:** Faculty can filter students in real-time by:

- Search bar (Student Name or Roll Number)
- Academic Department (Computer Science, Information Tech, ECE, Mechanical)
- Academic Year (1st, 2nd, 3rd, 4th Year)
- Target Career Domain (all 25 domains)
- Placement Certificate Status (Pending, Approved, Rejected, Not Eligible)

## C. Interactive Student Dossier Modal:

- Clicking any student in the table opens an in-depth **Student Dossier Modal** showing:

- Exact breakdown of Resume ATS score and attempt history.
- Technical Interview drill score vs HR Interview drill score.
- Gemini AI Mock Interview score and feedback.
- Qualitative reflections and total XP points earned.

## D. Certificate Governance & University Audit Export:

- **One-Click Faculty Endorsement:** Faculty can click **'Approve'** or **'Reject'** on a pending certificate directly from the dossier modal. The decision updates Supabase instantly, immediately unlocking the verified certificate in the student's portal.
- **CSV Accreditation Export:** One-click export downloads a structured CSV file for University Placement drives, NAAC/ABET accreditation audits, and institutional board meetings.

# 4. Technical Integrations & How Things Actually Work

| Integration | Implementation File | Exact Working Mechanism |
|---|---|---|
| **Supabase Cloud Database & Auth** | supabase.js<br>admin.js<br>router.js | Initialized via supabase.createClient(URL, ANON_KEY).<br>• **Auth:** supabase.auth.signUp() and signInWithPassword().<br>• **Database:** Table student_states stores full state JSON linked by user UUID.<br>• **Sync:** saveState() automatically debounces and upserts student progress into Supabase.<br>• **Admin:** Fetches all rows from student_states in real-time. |

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| **Google Gemini 3.6 Flash AI** | js/ai.js | Uses endpoint: https://generativelanguage.googleapis.com/v1beta/models/gemini-3.6-flash:generateContent?key=...<br>• **Technical & HR Interviews:** Generates 5 dynamic MCQs with explanations in strict JSON.<br>• **Mock Interview:** Evaluates written answers, returns exact 4-tier scores.<br>• **Anti-Gibberish:** Checks vowel ratio < 12% and assigns 0% to nonsense. |
| **Reactive Single Page Router** | js/router.js | Listens to window.onhashchange. Reads window.location.hash (e.g. #/dashboard), maps it to a view function in js/student.js (e.g. screenDashboard()), and injects HTML into #app without reloading the browser page. |
| **3D Canvas Warp-Speed Engine** | js/student.js (Resume Builder) | Pure HTML5 Canvas 2D context using 3D perspective projection formula: screenX = (x3D / z3D) * fov + centerX. 110 depth coordinates drift towards the camera. Typing in input fields triggers an event listener that boosts velocity from 1.0 to 10.0, rendering cosmic speed rings. |
| **Domain Workstation Visualizer** | js/ai.js<br>js/student.js | Local asset mapping engine in DOMAIN_IMAGE_MAP connects all 25 domain IDs directly to high-definition 16:9 images in img/ (e.g. aiml -> AI&ML.png;). Loads in 0ms with smooth CSS opacity transition and onerror fallback. |

## 5. Top 10 Exam & Viva Questions (With Model Answers)

**Q1: Why did you build this as a Vanilla JS Single Page App instead of using React or Angular?**

Answer: Vanilla JavaScript offers zero bundle overhead, sub-millisecond DOM execution, and complete control over rendering cycles without heavy external dependencies. It proves a deep foundational mastery of core web standards (DOM APIs, asynchronous event loops, Canvas manipulation, and hash routing). For enterprise and collegiate scale, it loads instantly even on low-bandwidth networks.

**Q2: How does the Readiness Index calculate a student's score?**

Answer: The Readiness Index is a weighted composite algorithm (0–100). It synthesizes 4 core competencies: Career Fit Assessment score (15% weight), Resume ATS quality score (25% weight), Technical Interview drill score (30% weight), and Gemini AI Mock Interview performance (30% weight). Students must cross 75% to become eligible for placement endorsement.

**Q3: How do the Student Portal and Admin Portal stay synchronized?**

Answer: They share a unified cloud backend powered by Supabase PostgreSQL. When a student completes an assessment or interview, saveState() writes their state JSON to the student_states table. When faculty open admin.html, it queries the student_states table in real-time. When faculty approve a certificate in Admin, the status changes to 'Approved' in Supabase, which immediately updates the student's screen on their next refresh or action.

**Q4: Which Google Gemini model are you using, and why?**

Answer: We use models/gemini-3.6-flash. Earlier versions like gemini-2.5-flash and gemini-1.5-flash were deprecated by Google for new users (returning 404). Gemini 3.6 Flash provides sub-second latency, strict JSON adherence, and the lowest credit consumption per query, making it ideal for high-volume academic interview simulations.

**Q5: How does your system prevent students from cheating the AI Mock Interview with gibberish?**

Answer: We implemented a dual-layer anti-gibberish filter. First, client-side function isGibberishText() inspects letter-to-vowel ratios. If the vowel ratio is below 12% in words over 6 letters (e.g., 'sfdfdsfsnfbsjdfsdjf') or has repeated consonants, it immediately assigns 0% across all metrics without wasting API tokens. Second, Gemini's prompt explicitly instructs that meaningless keyboard mash must receive 0 scores for Relevance, Communication, Technical Depth, and Structure.

**Q6: How does the 3D flight tunnel in the Resume Builder work?**

Answer: It is rendered on an HTML5 Canvas using pure 3D-to-2D perspective trigonometry. 110 active star coordinates have X, Y, and Z values. The projection formula screenX = (x3D / z3D) * FOV + centerX calculates where each point appears. Typing into any form input fires an event listener that surges velocity from 1.0 to 10.0, simulating a sci-fi warp-speed travel effect.

**Q7: How did you implement real-time question generation in Technical and HR interviews?**

Answer: When a student starts an interview, generateInterviewQuestionsWithAI() sends a single compact prompt to Gemini 3.6 Flash requesting 5 domain-tailored questions in strict JSON format. For Technical, Gemini acts as a Principal Engineer; for HR, it acts as a Senior HR Director. Each question includes 4 choices, the correct index, and a recruiter rationale explanation.

**Q8: What happens if the Gemini API is down, rate-limited, or internet drops?**

Answer: The system has built-in offline fault tolerance. If fetch() fails or returns non-200 status, generateInterviewQuestionsWithAI() catches the error and smoothly returns curated domain fallback questions from INTERVIEW_QUESTIONS. The UI never freezes or throws unhandled promise errors.

**Q9: What database schema are you using in Supabase?**

Answer: We use a relational schema in PostgreSQL 15. The core table is public.student_states, where id (UUID) references auth.users(id), with columns student_email (text), state_data (jsonb storing all scores, badges, attempts, and profile info), and updated_at (timestamptz).

### Q10: What are the future enhancements you plan for CareerForge AI?

Answer: Future enhancements include real-time voice-to-voice interview conversation using Gemini Live Audio WebSockets, direct ATS resume parsing via PDF OCR extraction, and automated campus placement drive scheduling integrated with college ERP systems.